Magnetostratigraphic Study of the Continental Lower Miocene Suchilquitongo Formation, Southern Mexico

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ABSTRACT

New paleomagnetic data from southern Mexico provide a magnetostratigraphic framework to correlate mammal-bearing continental sections of the Suchilquitongo Formation to the Early Miocene part of the geomagnetic polarity time scale. A total of 127 paleomagnetic samples were collected from 40 horizons belonging to two sections. The sequence is characterized by a succession of cream to light pink to gray, thin- to medium-bedded tuffaceous sandstones and siltstones, partly silicified limestones, and a dominantly light green ignimbrite deposited in a fluviolacustrine environment. Remanent magnetization generally has been characterized by two components, a present-day field component with low-unblocking temperatures up to 350°C and a characteristic component with unblocking temperatures up to 580°–610°C. The characteristic component is interpreted as primarily based on antipodal populations of remanence data. Six sites at the top of the section (inclination = 40.5°, declination = 11.3°, α_{95} = 1.4, and K = 471) have a mean paleomagnetic direction rotated clockwise about 17° with respect to the expected Miocene direction derived from the North American polar wander curve (inclination = 354°, declination = 30°). According to biochronologic age, the magnetostratigraphic results from the mammal-bearing sedimentary-type section in the Suchilquitongo area can be correlated with chron C5Dn-C5Cn3n at approximately 17.61–16.55 Ma. An age of about 17 Ma is proposed for the fossiliferous level, corresponding to an Early Miocene age.

Introduction

Fossil mammal studies are among the most efficient tools for dating and correlation of nonmarine Tertiary successions (Prothero 1995; Woodburne and Swisher 1995; Opdyke and Channell 1996). Nevertheless, the fossil mammal record is usually restricted to either isolated horizons or lenses in stratified sequences. This discontinuous record has often hampered the use of the standard biostratigraphic approach that has been extensively developed both in North America and in Europe, where the Paleogene and Neogene mammal biochronological subdivisions have been revised in the past decades.

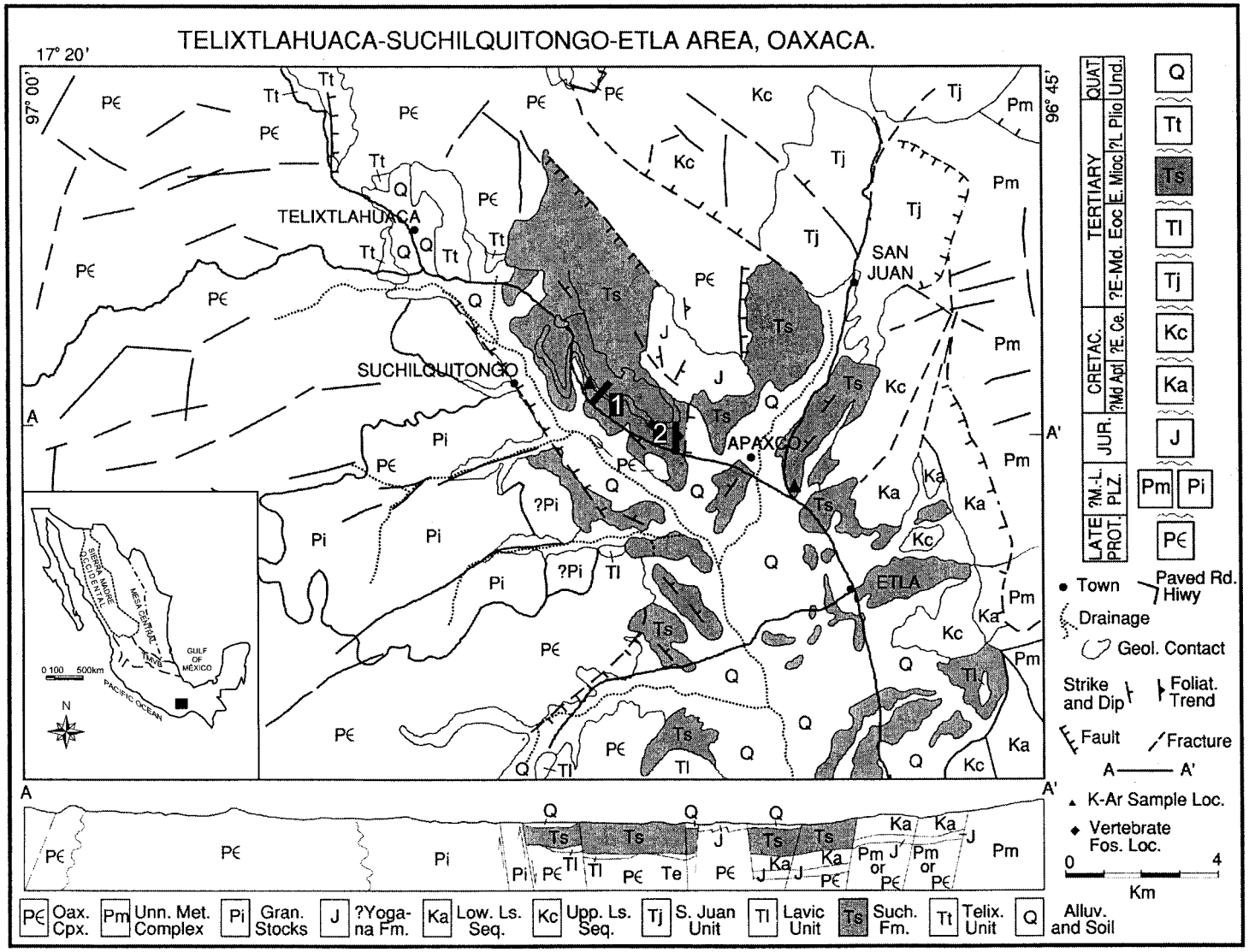
The North American land mammal age (NALMA) system is one of the most robust and long-standing paleontologic time scales. It was first formalized by Wood et al. (1941, p. 5) as a series of "provincial land-mammal ages." This system has become the benchmark for correlations in the terrestrial Cenozoic record of North America and a source of inspiration for analogous time scales dealing with other continents (Mein 1975). In recent years, the system has been revised heavily (Woodburne 1987; Prothero 1995; Woodburne and Swisher 1995; Kowallis et al. 1998), not just because of continuing empirical discoveries but because mammalian paleontologists have shifted to an incommensurate paradigm. After a century of allowing the time scale to be flexible, informal, and grounded on faunal correlations that took all the evidence into consideration, efforts have been made to transform it into a formal chronostratigraphic system.

During a paleontologic reconnaissance of the Neogene Suchilquitongo Formation in southern Mexico, a biochronologic and radiometric age of 16.5 ± 0.3 Ma and 17.4 ± 0.3 Ma was established with some disagreement (Ferrusquia-Villafranca 1990*b*). Magnetic polarity sequences of mammal-bearing sections will eventually result in the calibration of fossil localities to the geomagnetic po-

Manuscript received June 18, 2002; accepted March 18, 2003.

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larity time scale and thus assigning absolute ages to characteristic fauna assemblages. To make a direct correlation of the Suchilquitongo fauna assemblages to the geomagnetic polarity time scale (GPTS), we sampled two sections from this area that play a key role in the Neogene mammal sequence of Central America. In this article, we present new magnetostratigraphic results, the correlation to the GPTS, and comparison of our results with the K-Ar isotopic age dating and with very preliminary paleomagnetic data (Urrutia-Fucugauchi and Ferrusquia-Villafranca 2001).

Geological Setting and Sections

The general Tertiary history of Mexico has long been known to include volcanic events accompanied by the accumulation of thick continental sedimentary sequences. With these continental sedimentary sequences are found fossil mammal localities (Ferrusquia-Villafranca 1975, 1978, 1990*a*, 1990*b*, 1990*c*; Miller and Carranza-Castañeda 1984; Carranza-Castañeda 1988, 1992; Carranza-Castañeda and Miller 2000). Correlation of the fossiliferous localities to the NALMA has been mostly based on biochronological dating. However, the presence of volcanic material has allowed isotopic age dating and greatly improved correlation of some of these localities (Kowallis et al. 1998; Carranza-Castañeda and Miller 2000).

The area studied is located in the central part of Oaxaca state (fig. 1), about 20 km NW of Oaxaca city. Wilson and Clabaugh (1970) published a geologic map of the area. They described the Tertiary sequence formed by sedimentary and volcanic rocks and proposed the name Suchilquitongo Formation. According to these authors, the Tertiary sediments lie unconformably on rocks of Precambrian and Cretaceous age and, by fault contact, with rocks of Jurassic age. These sediments are cut by faults trending 45° NW and 65° SW, partitioning the Tertiary into several tilted (8°–30°) blocks in distinct directions. The basal Tertiary rocks differ from place to place; at the section-type area, a volcanic flow rests directly on Precambrian, whereas on a hill to the north of Suchilquitongo town, a basal conglomerate with limestone pebbles lies between a volcanic flow and Precambrian schist. The Suchilquitongo Formation lies above a plagioclase-rich basalt; the latter covers the lower conglomerates of this formation. This formation comprises a succession of light-colored tuffaceous silts, partly silicified limestones, tuffs, clays, and an ignimbrite that is exposed between the towns of Huitzo and Etla. Etla is built on the light green ignimbrite; for this reason, the ignimbrite is named the Etla Ignimbrite member of the Suchilquitongo Formation. The presence of mammal fossils in volcaniclastic beds allowed Wilson and Clabaugh (1970) to assign this formation a Late Miocene age.

The sampled sections of the Suchilquitongo Formation have the similar lithologic facies, sometimes with lateral lithology variations. They consist mainly of sandstones, claystones, lacustrine limestones, conglomerates, ignimbrites, and volcanic ash tuffs (fig. 2). Throughout the area, brown and green ignimbrite deposits have been extensively quarried. The possible source of the ignimbrites has not been identified. Similar ignimbrite exposures are found near Yagul and Mitla, SE of Oaxaca city. The sections are capped by a thick, unconsolidated conglomerate with igneous boulders and cobbles up to 7 cm in diameter interbedded with thin sandstone beds. This conglomerate is thought to be of Pliocene age (Wilson and Clabaugh 1970) and suggests a period of active tectonism in the surrounding area at the time of deposition.

Paleomagnetic Sampling and Measurement

Following standard procedures, we collected either hand samples or a minimum of three drilled core samples, oriented in situ using a magnetic compass. The average vertical separation between sites is about 2 m or less, and there are no gaps of more than 10 m in the sampled sections, except in the unconsolidated conglomerate of the upper part of section 1, which is not favorable for sampling (fig. 2). Strike and dip of the strata were measured for tilt correction of the remanent magnetic direction. More than 127 samples from two sections were collected from 40 sites. All samples were cut to the

Figure 1. Simplified schematic geologic map for the Etla sector of Oaxaca Valley, central Oaxaca, showing sampling sections (vertical sections marked by *1* and *2*). The filled squares and roman numerals are paleomagnetic sites sampled by Urrutia-Fecugauchi and Ferrusquia-Villafranca (2001). $P\in$ = Oaxaca complex; *J* = Yogana Formation; *Ka* = Middle Cretaceous sequence; *Kc* = Late Cretaceous sequence; *TI* = polymictic conglomerate; *Q* = Quaternary deposits; *1* = town; *2* = drainage.

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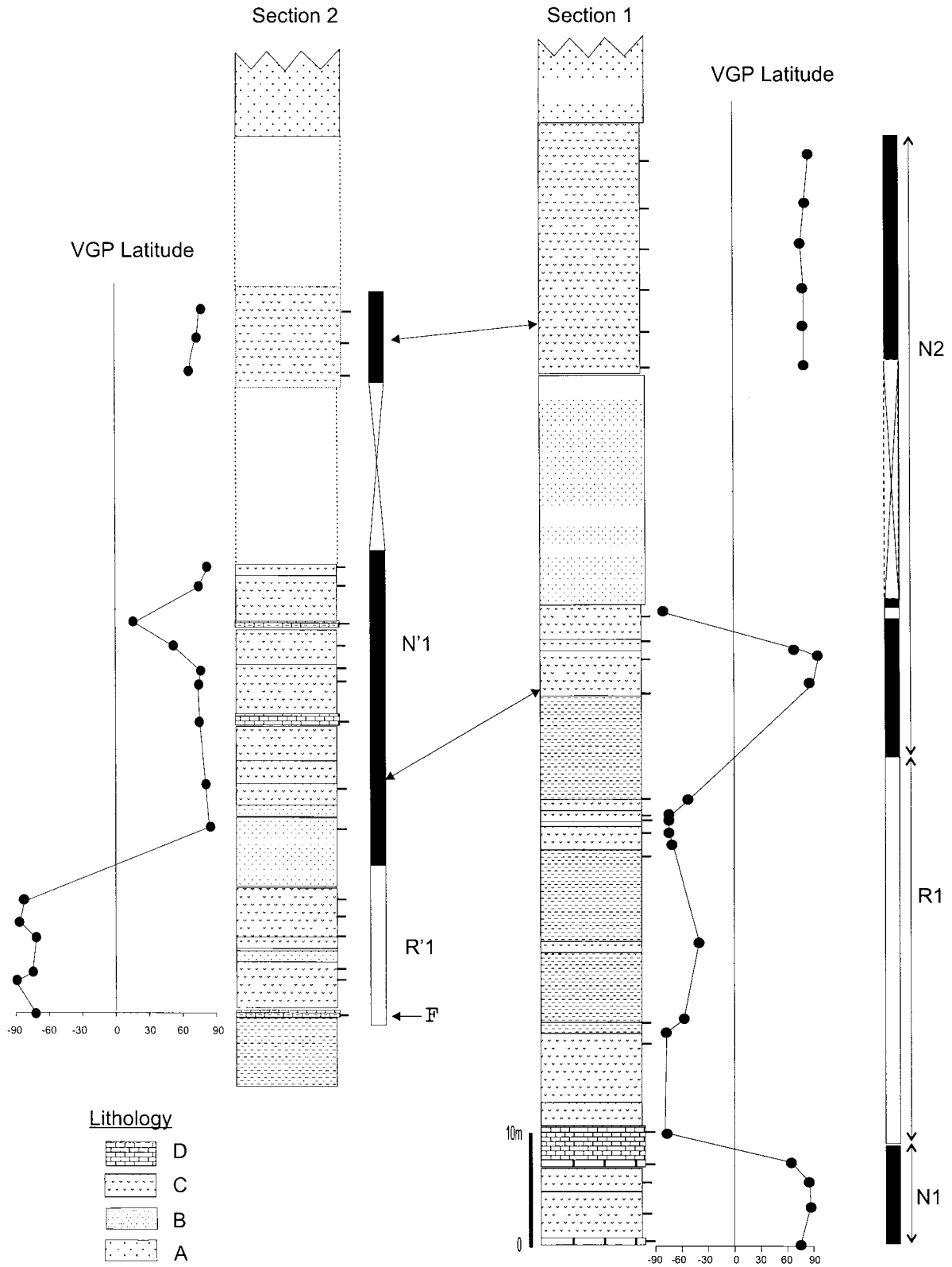


Figure 2. Lithology, magnetostratigraphy, and stratigraphic position of sampling sites of sections 1 and 2. Positive and negative virtual geomagnetic pole (VGP) latitudes represent normal and reversed polarities, respectively. *F* = location of fossil mammals. *A*, Conglomerate; *B*, siltstone; *C*, volcanic ashes/ignimbrite; *D*, limestone.

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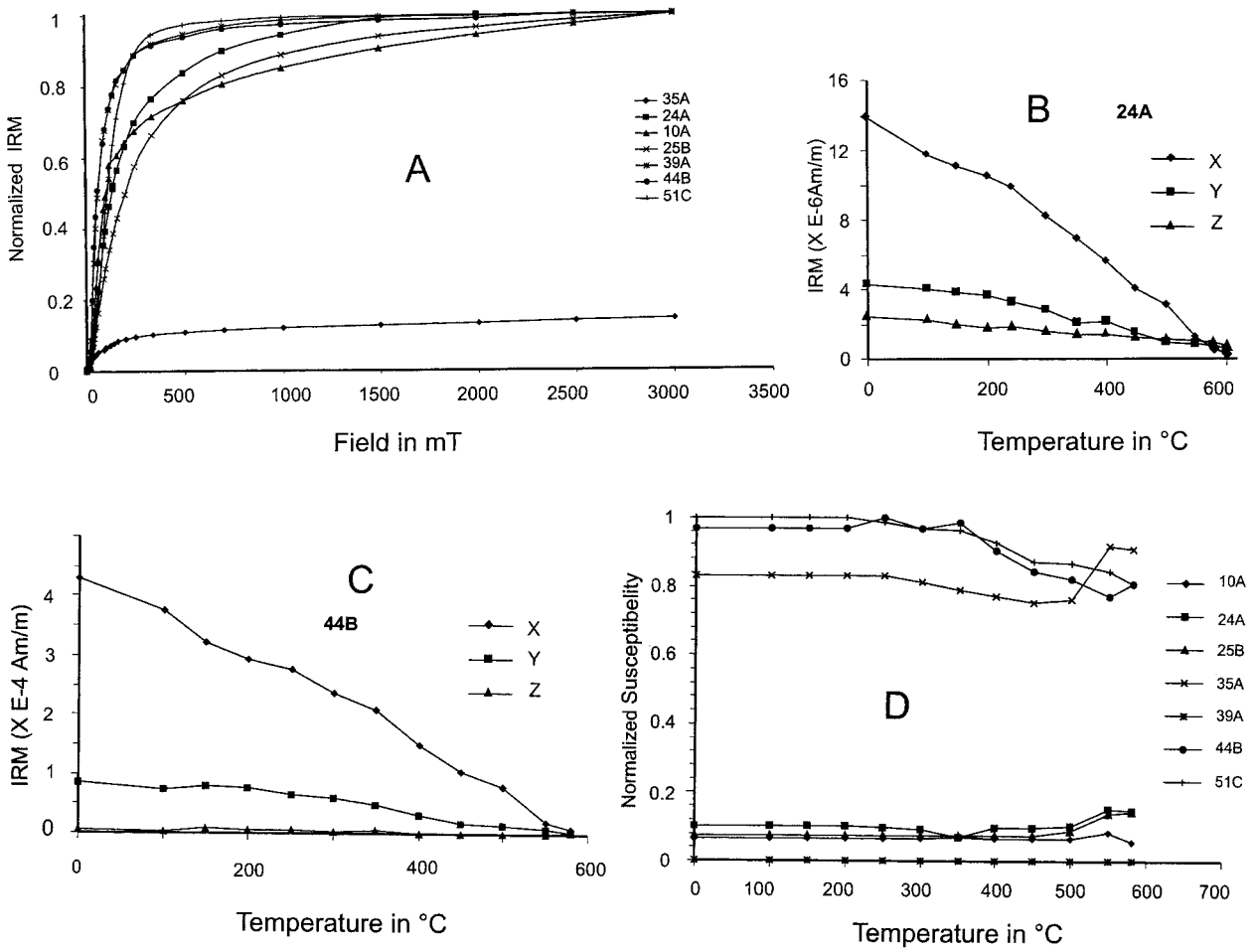


Figure 3. *A*, Isothermal remanent magnetization (IRM) acquisition curves of selected samples from both sections. *B*, *C*, Thermal demagnetization of IRM. *D*, Susceptibility values after each demagnetization step measured at room temperature for selected samples.

standard paleomagnetic size. The field relations and correlation of our two sections (3 km separated; fig. 1) are defined by important marker beds that can be traced from section to section, especially the ignimbrite and conglomerate at the top of the sections. Natural remanence magnetization (NRM) was measured using a JR5 spinner magnetometer. Alternating field (AF) and thermal demagnetizations were carried out using a Molspin shielded demagnetizer with a peak field of 100 mT and a shielded Schonstedt-type furnace.

Both visual analysis of stereographic and orthogonal plots (Zijderveld 1967) and statistical methods using principal component analysis (Kirschvink 1980) were used to determine the characteristic magnetization from demagnetization data. Mean directions were computed using Fisher's (1953) statistics. The magnetic mineralogy was investigated by studying the coercivity isothermal remanent magnetization (IRM) acquisition spectrum and the unblocking temperature points of a laboratory imposed triaxial IRM (Lowrie 1990).

Magnetic Mineralogy. We performed rock-magnetic experiments on samples of the different rock types present in the studied sections to investigate the minerals carrying the magnetization. These rock-magnetic experiments included IRM acquisition and subsequent thermal demagnetization. Stepwise increasing of direct fields up to 3T was applied to each of the samples to determine their IRM acquisition pattern. In general, samples showed an initial rapid acquisition of about half of the total IRM at fields of 120 mT, followed by a continuous slow rate with saturation up to the end of the experiment (fig. 3*A*).

Before thermal demagnetization of the IRM, the

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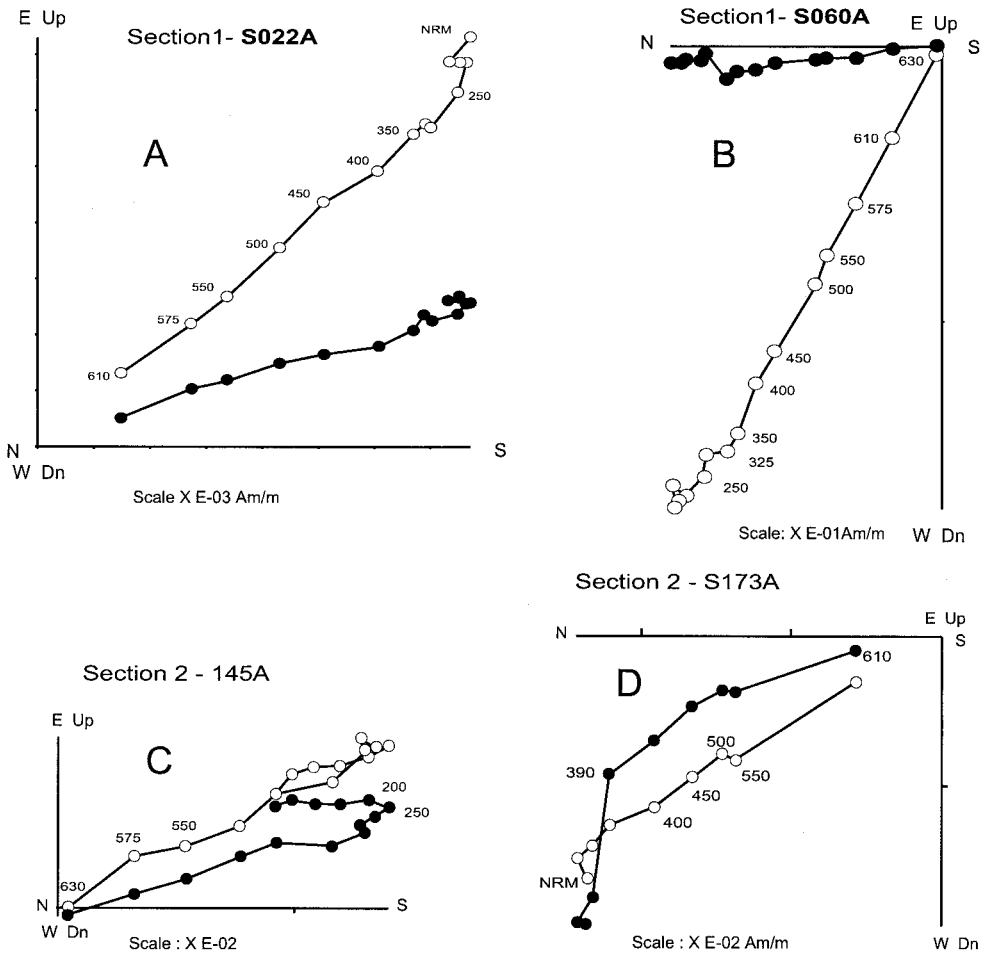


Figure 4. Representative Zijderveld diagrams of the Suchilquitongo Formation; *A*, *B*, section 1; *C*, *D*, section 2

samples were magnetized along three axes (*X*, *Y*, and *Z*) using the Lowrie method (1990). For all samples, we applied a weak field of 0.08 T along the *X*-axis, 0.5 T along the *Y*-axis, and 2 T along the *Z*-axis. The acquisition of these IRMs along the three axes has been carried out using a homemade coil impulse magnetizer.

After IRM acquisition magnetization, we applied the thermal demagnetization procedure. Magnetic susceptibility was measured after each heating step to detect possible mineralogic changes due to heating. In the diagrams (fig. 3), the demagnetization curves correspond to the components of the IRM according to the *X*-, *Y*-, and *Z*-axes. The diagrams obtained for the various samples showed similar behavior. Figure 3*B* and 3*C* shows an example of these results. The curves of the three components (*X*, *Y*, and *Z*) do not show any inflection point but decrease steadily until demagnetization is complete between 580° and 610°C. The susceptibility curve remains almost constant until the temperature range 350°–500°C, after which several small fluctuations occur (fig. 3*D*). The combined results from both IRM acquisition and thermal demagnetization are characteristic of magnetite and/or hematite, suggesting that it is the main magnetic carrier.

Natural Remanence Magnetization. The NRM intensity is generally high, ranging from 0.1 to 10 mA/m, and is strongly related to the lithology. Generally, the brown ignimbrite (at the top of the sections) and nonaltered ash deposits have higher intensities than do limestones and marls. Thermal and/or alternating field demagnetization of the samples reveals that the total NRM is generally composed of one or two components. A large secondary component removed at temperatures between 240° and 300°C shows approximately a present-day field direction before bedding tilt correction and is therefore of recent origin. A char-

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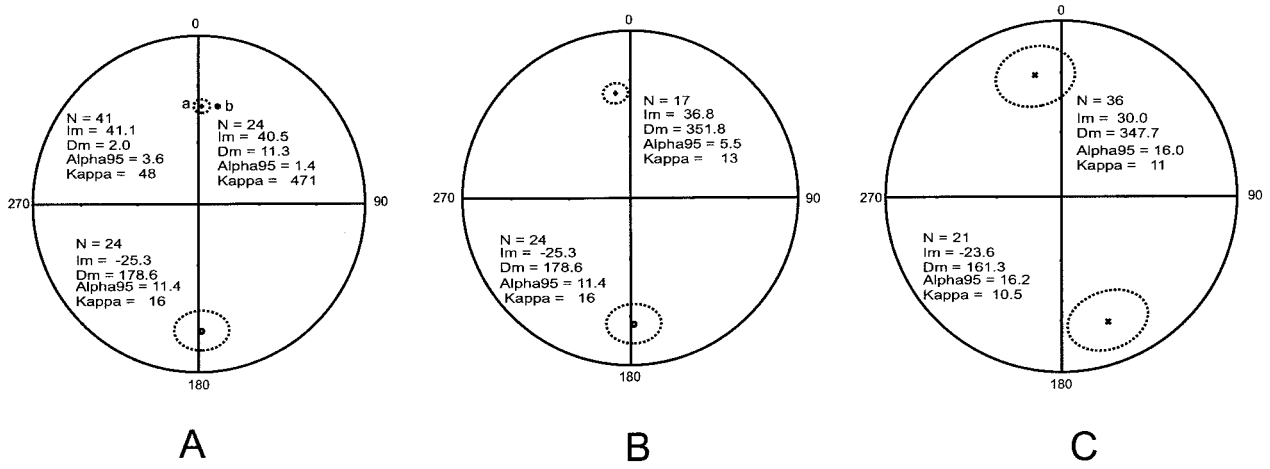


Figure 5. Equal-area stereographic projection of characteristic directions (normal and reversed polarities). *A*, Section 1; *a* = mean direction for all normal samples; *b* = sites exhibiting clockwise rotation. *B*, Mean direction excluding the six sites. *C*, Section 2. The direction of normal and reversed samples appear to be antipodal, and the reversal test is positive. The angle between normal and reversed directions is 8.8° and 12.5° for sections 2 and 1, respectively.

acteristic remanent magnetization is generally removed at temperatures between 580° and 630°C (fig. 4). This temperature is higher than the magnetite unblocking temperature and suggests the presence of hematite as the main magnetic carrier for this sample. Demagnetization vector plots are generally of good quality, and in most cases, characteristic directions can be reliably determined. Only limestones show less reliable results because of a very weak magnetic signal.

Statistical Treatment and Results. Data resulting from thermal and AF demagnetization were plotted on orthogonal vector plots (Zijderveld 1967). To determine characteristic magnetic directions, principal components analysis was carried out on all samples. These paleomagnetic directions were then analyzed, using Fisher statistics, to determine site mean declinations, mean inclinations, and associated precision parameters.

For section 1, the mean direction of normal sites is declination = 347.7, inclination = 30.06 (α = 16.0, K = 11), and the mean direction of reverse sites is declination = 161.34, inclination = −23.6 (α = 16.2, K = 16; fig. 5*A*). However, the site mean of the normal sites for section 2 is declination = 2.06, inclination = 41.14 (α = 3.6, K = 48), whereas the mean direction of all reverse sites is declination = 178.66, inclination = −25.34 (α = 11.4, K = 16; fig. 5*C*). The difference between the mean directions of the normal and reverse polarity populations for two sections may result from a possible incomplete removal of a secondary overprint or a local tectonic deformation. For samples of normal polarity, six sites (section 1) exhibit clockwise (CW) rotation of 17° (declination = 11.3°, inclination = 40.5°; fig. 5*B*) with respect to the expected Miocene direction derived from the North American polar wander curve at 20 Ma (declination = 354°, inclination = 30°; Harrison and Lindth 1982). Previous paleomagnetic results have shown that the ignimbrite has undergone rotation of about 15° (Urrutia-Fucugauchi and Ferrusquia-Villafranca 2001). Since there is no evidence of significant alteration in these samples, this group with CW rotation could have been produced by a local tectonic deformation, attested to by the presence of the horizontal faults in the ignimbrite and the conglomerate beds. Excluding these six sites, the overall mean of normal polarity sites for the Suchilquitongo data set is declination = 351.87°, inclination = 36.85° (N = 24, α = 5.5, K = 13). The direction of normal and reverse polarity sites (fig. 5*A*, 5*C*) are antipodal, and the reversal test is positive (McFadden and McElhinny 1990).

Magnetostratigraphy

The presence of normal and reverse polarities at a relatively high temperature (>300°C) suggests that the NRM of the samples is primary. We have examined the characteristic magnetizations to evaluate the origin and nature of the paleomagnetic record. Some samples were eliminated because of inconsistency of directions during the successive

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steps of demagnetization or because of their weak magnetization.

The virtual geomagnetic pole (VGP) latitude of the characteristic paleomagnetic directions yields a magnetic polarity sequence in both sections (fig. 2; table 1). In the lower part of section 1, results are of normal polarity. This polarity, N1, is represented by four successive sites. Above this is a 40-m-thick interval represented by eight sample sites that give reverse polarity, R1. The polarity zone N2 is represented by 10 sample sites, with a reverse polarity represented by only one site (samples 45 and 46); however, because the zone does not extend laterally to the other section, this polarity is probably an artifact. Section 2 exhibits two polarity intervals, R′1 represented by six sample sites and N′1 represented by 12 sample sites. In conclusion, this paleomagnetic study (composite section) highlights at least the succession of three magnetic polarity zones. Figure 2 depicts the position of paleomagnetic sites and interpreted magnetic polarity zones throughout the two studied sections.

Correlation. A variety of previous biochronology and geochronology studies on the Suchilquitongo Formation provides constraints for the correlation of the sections studied here. Two K-Ar dates were reported earlier for the Etla Ignimbrite member (16.5 ± 0.3 and 17.4 ± 0.3 Ma; Ferrusquia-Villafranca et al. 1974; Ferrusquia-Villafranca 1990*b*). Three new K-Ar dates obtained from mineral concentrates of samples from the Etla Ignimbrite have older values, 19.2 ± 0.5 to 20.6 ± 0.3 Ma (Urrutia-Fucugauchi and Ferrusquia-Villafranca 2001).

Since the original description of the Suchilquitongo Formation, Wilson and Clabaugh (1970) discovered a molar of *Merychippus* that allows this formation to be placed in the Late Miocene. Paleontologic work in the Suchilquitongo area has permitted a significant increase in knowledge of the Middle Miocene mammal faunas of southern Mexico (Ferrusquia-Villafranca 1975, 1978, 1990*b*). The previously known taxa identified by Ferrusquia-Villafranca (op. cit.) are as follows: *Plionictis oaxaquensis*, Rhinocerotidae gen. and sp. indet., cf. *Paratoceras* sp., *Merychyus* aff. *M. minimus*, *Oxydactylus* cf. *longipes*, *Gomphotherium* sp., and Archaeolaginae gen. and sp. indet. All of these fossils came from four closely spaced localities. Recently, Jimenez-Hidalgo (1998) and Jimenez-Hidalgo et al. (1999) noted the presence of an Antilocapridae genr. and sp., indet and ?tylopodan artiodactyle family with molarized permanent premolars. On the basis of the resemblance of some species with those known in North America, Ferrusquia-Villafranca (1990*b*) assigned the Suchilquitongo Formation to the latest Hemingfordian–earliest Barstovian, according to the land mammals biochronology established in North America. In the absence of large mammals used to define faunal-age boundaries for the Barstovian, it has been difficult to differentiate clearly late Hemingfordian and early Barstovian faunas in the Suchilchitongo Formation. The absence of proboscidian as well as *Hemicyon* and *Pseudocyon*, whose first record define early and late Barstovian, show that this formation is older than the Barstovian. Ferrusquia-Villafranca (1984) indicated that the

Table 1. Paleomagnetic Data for Suchilquitongo Formation

Site	n/N	Mean direction (tilt corrected) Declin.	Inclin.	α_{95}	k	VGP latitude
Section 1:						
1	1/3	345.11	40.54	…	…	74.7
2	4/4	4	36.11	8.7	112	85.1
3	2/2	7	29.09	…	…	83.1
4	1/2	26	47.41	…	…	63.4
5	3/3	170.06	−28.73	18.4	46	−80.3
6	2/2	210.73	−22.31	…	…	−59.8
7	2/2	128.37	−31.08	…	…	−40.7
8	3/3	196.14	−46.84	12.3	101	−72.1
9	3/3	164.10	−33.87	10.1	149	−74.8
10	4/4	198.12	−45.36	8.8	110	−75.4
11	1/2	187.64	−11.79	…	…	−76.7
12	4/4	165.99	−22.26	6.5	204	−63.7
13	1/2	6.14	46.84	…	…	77.5
14	3/3	360	34.98	22.5	31	87.7
15	2/2	341.05	9	…	…	67.6
16	2/2	178.11	−43.98	…	…	−81.0
17	5/5	10.32	42.3	4.7	270	77.8
18	3/3	11.12	41.65	2.7	2049	77.4
19	4/4	11.81	41.81	3	924	76.8
20	5/5	14.34	39.08	3	921	75.6
21	4/4	9.55	37.84	4.4	428	80.0
22	3/3	10.00	37.78	7.0	311	79.7
Section 2:						
1	4/4	160.13	−27.94	6.9	127	−70.7
2	4/4	179.74	−30.72	6.4	204	−89.4
3	3/4	164.64	−34.22	16.9	54	−75.2
4	3/3	168.39	−50.97	40.5	10	−71.9
5	2/4	177.65	−31.14	…	…	−87.7
6	5/5	173.6	−27.3	6.7	132	−83.3
7	2/3	355.87	29.9	…	…	85.92
8	3/3	8.19	38.52	28.1	20	80.9
9	3/3	349.10	12.45	8.1	94	74.8
10	2/2	7.3	9.22	…	…	75.7
11	3/3	344.79	26.74	5.4	518	75.0
12	3/3	339.0	26.98	22.9	30	52.4
13	2/3	277	70.81	…	…	17.8
14	2/2	355.1	50.76	…	…	74.8
15	1/3	351.24	37.73	…	…	80.7
16	3/3	347.42	56.62	28.5	20	66.9
17	3/3	353.1	51.9	19.5	41	73.2
18	4/4	355.51	49.56	4.7	381	75.9

Note. VGP = virtual geomagnetic pole.

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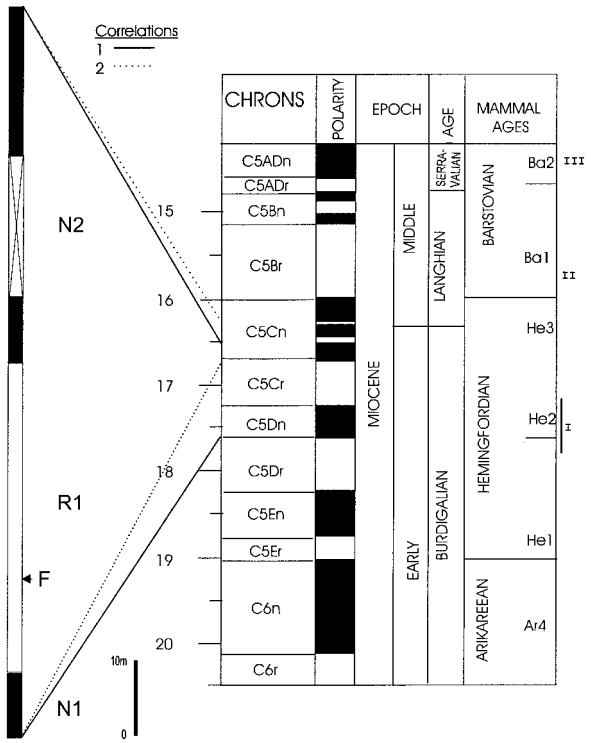


Figure 6. Two potential correlations of the composite section to the geomagnetic polarity time scale of Berggren et al. (1995), given relevant biochronologic and radiometric constraints. Correlation *1* places the Suchilquitongo Formation in chron C5Dn-C5Cn3n (17.61–16.55 Ma) with a sedimentation rate of about 7.2 cm/ky; correlation *2* places it in chron C5Cn3n-C5Cn2n (16.72–16.32 Ma) with a sedimentation rate of 10 cm/ka. The North American land mammal age boundaries and the dispersal taxa to the right follow Woodburne and Swisher (1995). *I*, *Petauristodon*, *Blackia*, *Eomys*, *Mioictice*, *Plionictis*, *Sthenictis*, *Mionictis*, and *Pseudaelurus*; *II*, *Hemicyon* and Proboscidean Datum (NW and SE North America); *III*, *Pseudocyon* and Proboscidean Datum (Interior North America).

presence of *Gomphotherium* sp. perhaps is a misidentification (I. Ferrusquia-Villafranca, pers. comm.).

Woodburne and Swisher (1995) correlate the land mammals of North America with the GPTS and identified 10 distinct mammal dispersal events during the Cenozoic era. The events recognized in the early and late Hemingfordian mammal age are the most impressive. The late Hemingfordian (He2) is defined by the occurrence of first immigration of *Petauristodon*, *Blackia*, *Eomys*, *Mioictice*, *Plionictis*, *Sthenictis*, *Mionictis*, and *Pseudaelurus*. If we consider only the well-defined species as *Plionictis*, this formation could belong to the Middle or Late Hemingfordian, that is, equivalent to an Early Miocene age.

Temporal Duration. If the studied sections were adequately sampled for magnetostratigraphy, then the number of reversals encountered can provide an estimate of the amount of time represented by the section (Johnson and McGee 1983). Using the equation from Johnson and McGee (1983), the duration Δt is calculated as follows: $\Delta t(\text{m.yr.}) = S\tau N$; where $S = -\ln(1 - 2p)/2$; τ = individual duration (time length) of polarity zones; N = number of local paleomagnetic sites sampled; $p = R/(N - 1)$, where R is the number of reversals encountered. For the section 1 sampled in this study, $N = 22$, and thus $p = 0.13$ and $S = 0.15$. Using the GPTS of Berggren et al. (1995), 10 polarity zones occur between 19.04 and 16 Ma, yielding a mean interval length (τ) of 0.31 m.yr. Substituting these values, we estimated that time required for deposition of this section is 1.12 m.yr.

The composite section of the Suchilquitongo Formation consists of three magnetozones (N1–N2) of which R1 is the longest. In conjunction with the consideration that the composite section should fall between 19.04 and 16 Ma and span 1.12 m.yr., there are only two potential correlations to the GPTS (fig. 6). The first possibility is to correlate the sequence with chrons C5Dn–C5Cn3n, which span an age from 17.61 to 16.55 Ma. We consider that this represents the best-supported hypothesis, but we also realize that it is not the only possible correlation. A second possibility is that the Suchilquitongo composite section could be correlated to chrons C5Cn2n–C5Cn3n. This correlation would require rejecting of the estimated duration (1.12 m.yr.) calculated above in favor of a shorter time span of about 0.4 m.yr. (which seems less likely). If the first proposal (correlation 1) is correct, we can calculate the sedimentation rate and propose an age for the fossiliferous level; the polarity R1 is complete, has a 40-m thickness, and is correlated with chron C5Cr, which lasts 0.55 m.yr. Consequently, its sedimentation rate is about 7.2 cm/k.yr. According to the correlation between both sections, the fossiliferous level is about 23 m above the lower boundary of the R1 polarity and therefore is at 17 Ma. Taking into account the sedimentation rate calculated above, the ignimbrite dated situated 80 m below the fossiliferous level (Wilson and Clabaugh 1970) should be at about 18.1 Ma. This age is younger than the K-Ar radiometric dating of 19.2 ± 0.5 to 20.6 ± 0.3 Ma of Urrutia-Fucugauchi and Ferrusquia-Villafranca (2001). This difference in age could be due to an underestimation of the

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measured thickness, since the age determined is based on this.

Conclusions

We have established a detailed magnetostratigraphy for two sections of lower Miocene age Suchilquitongo Formation. Paleomagnetic analysis of these sections of fluviolacustrine sediments from southern Mexico provide a paleomagnetic framework to correlate the Suchilquitongo Formation to the GPTS. Thermal demagnetization provided a nearly antipodal distribution of remanent directions consistent with the expected Miocene direction for this area. According to geochronologic and biochronologic constraints, two potential possibilities of correlation for the resulting polarity record to the GPTS are presented. Correlation 1 places the Suchilquitongo Formation within chron C5Dn-C5Cn3n, between 17.61 and 16.55 Ma, and an age at about 17 Ma is proposed for the fossiliferous horizon. Our new proposed age is older than that assumed in a previous study by Ferrusquia-Villafranca (1990*b*) and corresponds to the Early Miocene time.

ACKNOWLEDGMENTS

This article was developed in the framework of Research Projects B126 and DAGPA IN-102897 of the Instituto de Geofisica, Universidad Nacional Autonoma de Mexico. We thank B. J. Swisher and the anonymous reviewer for contributing greatly to improving the quality of the manuscript and I. Ferrusquia-Villafranca for providing much useful information.

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